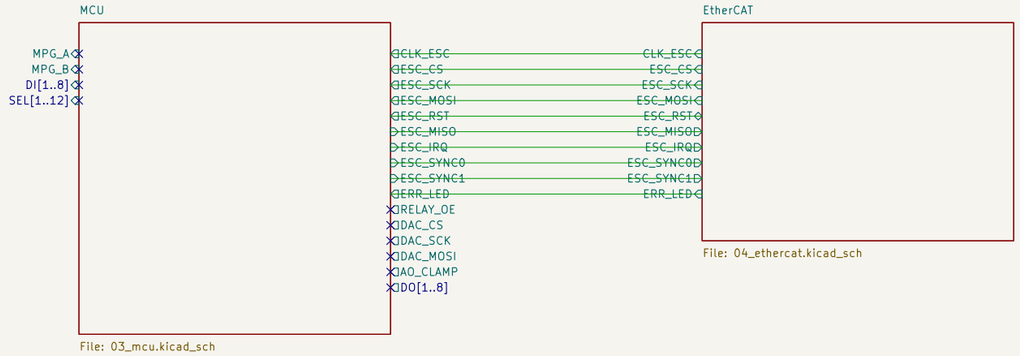
LCEC-IO — EtherCAT slave I/O board

Root sheet, milestone M4: Power (2), MCU (3) and EtherCAT (4). Sheets 5 to 8 (digital inputs, MPG, relay outputs, analog outputs) are added in M5; the full root sheet follows in M6.





No-connect flags on the MCU sheet pins are placeholders: they are replaced by wires to sheets 5 to 8 in M5.

Root sheet: Power, MCU and EtherCAT sheets

Yantra Innovation — LCEC-IO

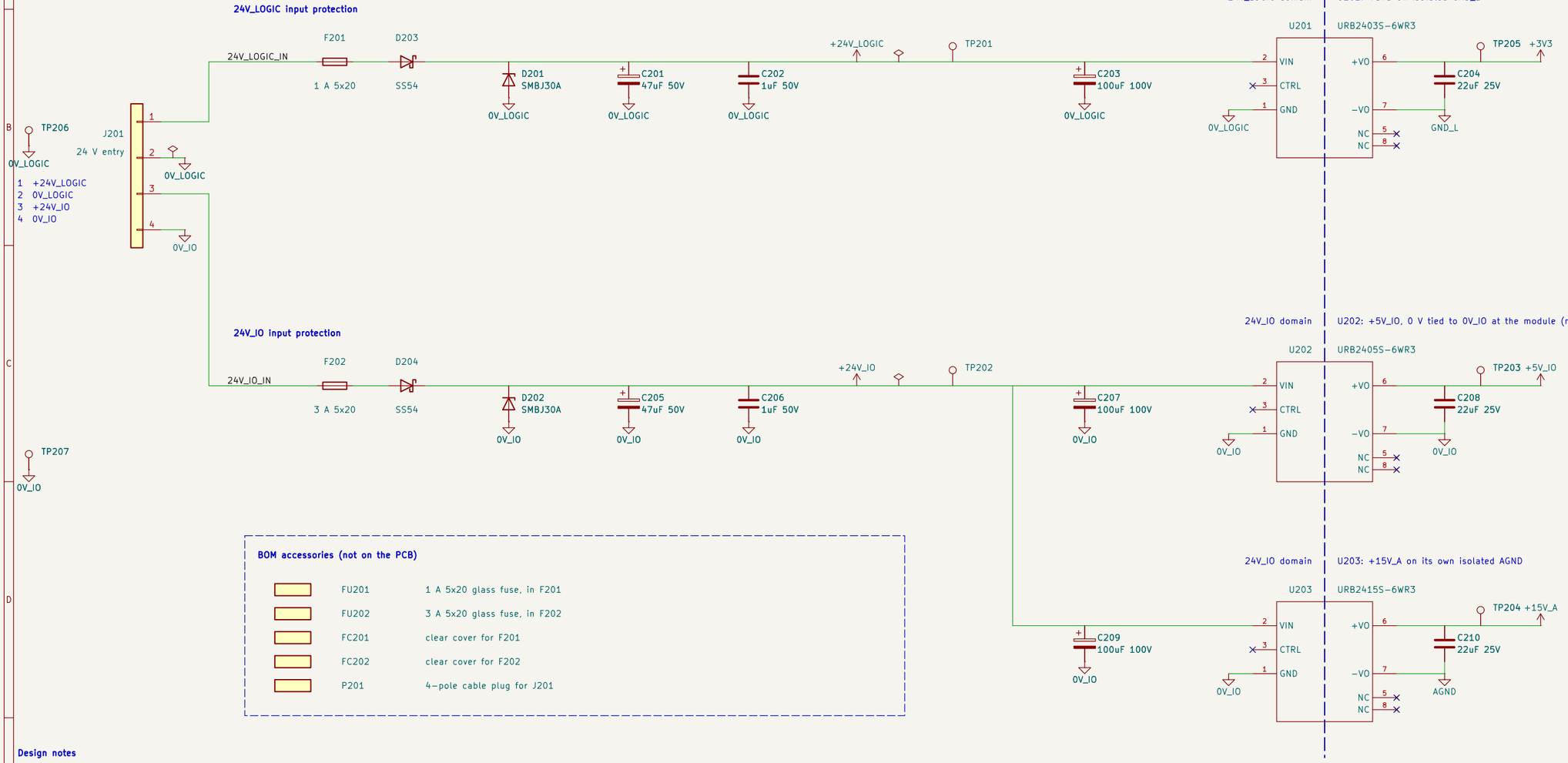
Sheet: /
File: lcec-io.kicad_sch

Title: LCEC-IO — EtherCAT slave I/O board

Size: A3	Date: 2026-11-06	**Rev: A**
KiCad E.D.A. 10.0.8		Id: 1/4

2 POWER – 24 V entries, protection, three isolated DC–DC modules

Input 24 VDC +/–20 % = 19.2 … 28.8 V. Per input: fuse > SS54 reverse–polarity diode > SMBJ30A TVS > 47 uF + 1 uF, then a 100 uF/100 V input capacitor at each URB module (URB datasheet Cin).



BOM accessories (not on the PCB)

	FU201	1 A 5x20 glass fuse, in F201
	FU202	3 A 5x20 glass fuse, in F202
	FC201	clear cover for F201
	FC202	clear cover for F202
	P201	4–pole cable plug for J201

Design notes

Loads: +3V3 about 0.4 A of the module's 1.35 A (30 %). 24V_LOGIC input about 0.3 A at full module load: 1 A fuse.

24V_IO: 5 V module 7.3 W + 15 V module 7 W at full load = 0.75 A, plus relay coils 0.21 A and sensors through the PTCs 1 A: 3 A fuse.

Input inrush I2t into the input capacitors stays below 1.1 A2s; the fuses are rated 4.26 A2s (1 A) and 12.5 A2s (3 A).

URB input 9–36 V (40 V continuous, 50 V for 1 s). SMBJ30A: 30 V stand–off above the 28.8 V maximum input, clamps 48.4 V at 12.4 A.

CTRL left open = module on. NC pins are not connected inside the module.

Two 24 V entries, protection, three isolated DC–DC modules

Yantra Innovation – LCEC–IO

Sheet: /Power/

File: 02_power.kicad_sch

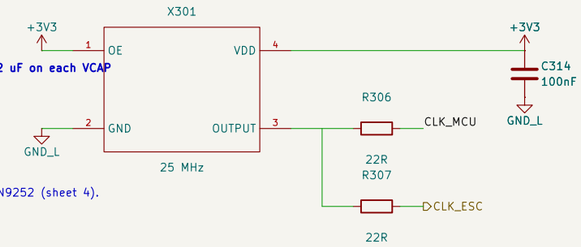
Title: LCEC–IO – 2 Power

Size: A3	Date: 2026–11–06	**Rev: A**
KiCad E.D.A. 10.0.8		Id: 2/4

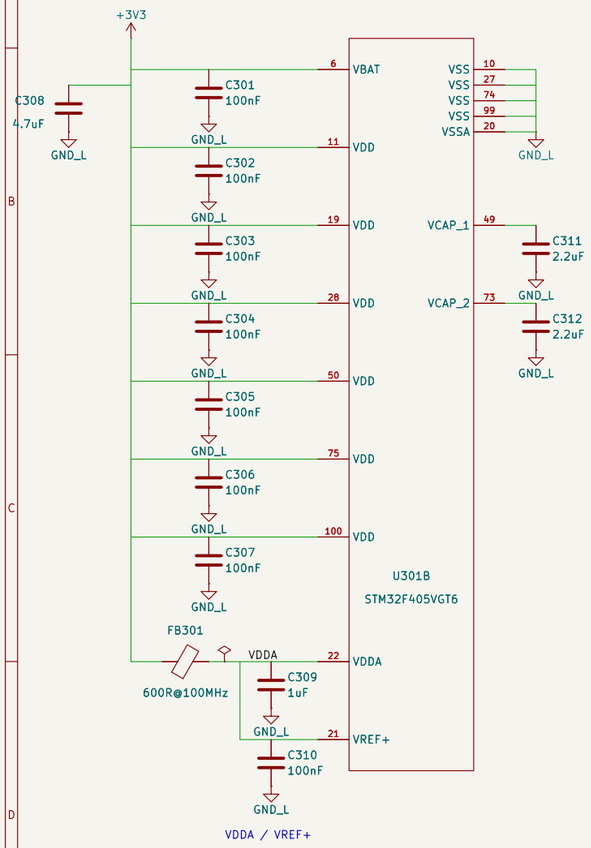
3 MCU — STM32F405VGT6, shared 25 MHz clock, SWD, UART2, heartbeat LED

51 signals used (pin map in docs/plan.md), PH1 is a test pad, the 30 pins of ports D and E are free for future use (no-connect flags). Firmware: HSE bypass, PLLM 25 / PLLN 336 / PLLP 2 = 168 MHz, IWDG enabled in option bytes.

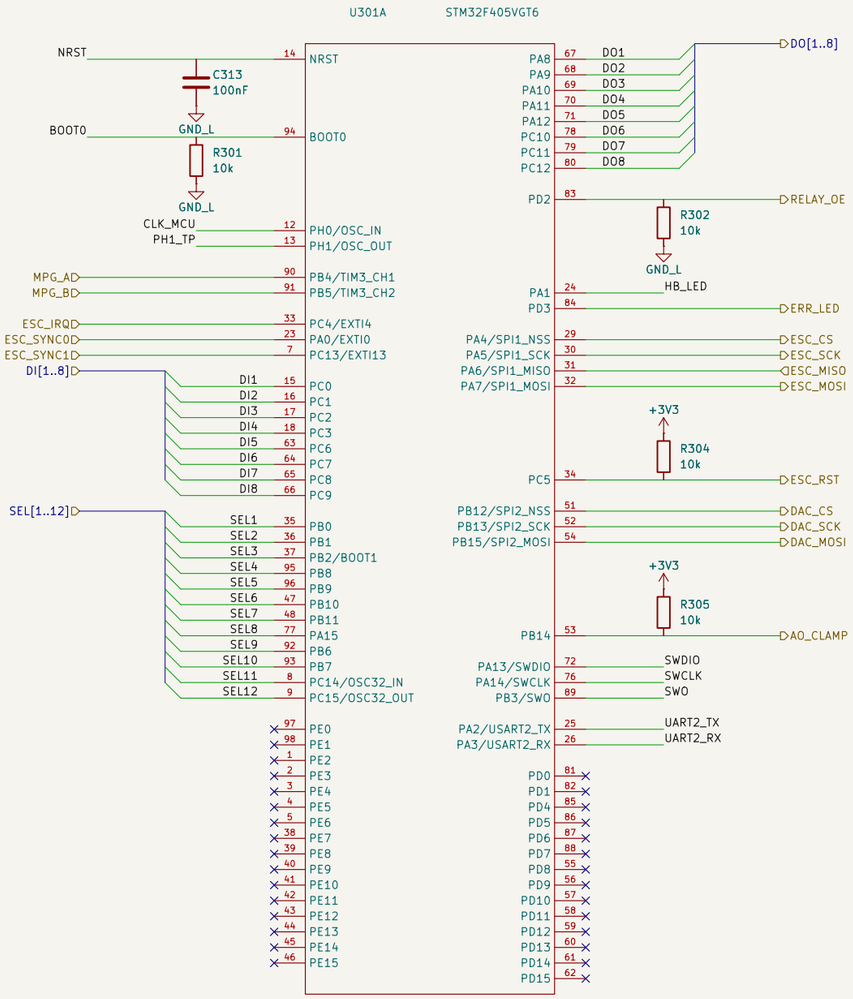
U301 power: 100 nF beside each VDD pin and VBAT, 4.7 uF bulk, VDDA through a ferrite, 2.2 uF on each VCAP



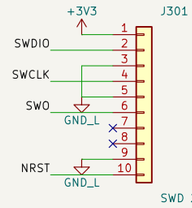
OE tied high = oscillator running. 22 R series resistors to PH0 (HSE bypass) and to the LAN9252 (sheet 4).



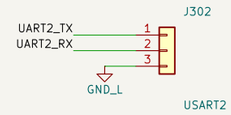
VDDA / VREF+



SWD (Cortex 10-pin): 1 VTref, 2 SWDIO, 3/5/9 GND, 4 SWCLK, 6 SWO, 7/8 not used, 10 NRST



USART2: 1 TX (PA2), 2 RX (PA3), 3 GND



Test points: SPI1 (LAN9252), SPI2 (DAC), relay OE, analog clamp, MPG A/B and the spare pin PH1





STM32F405VGT6 (LQFP-100), shared clock, SWD, UART, heartbeat LED

Yantra Innovation — LCEC-IO

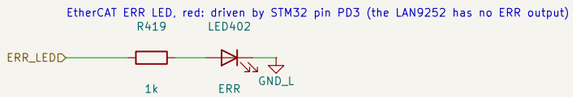
Sheet: /MCU/
File: 03_mcu.kicad_sch

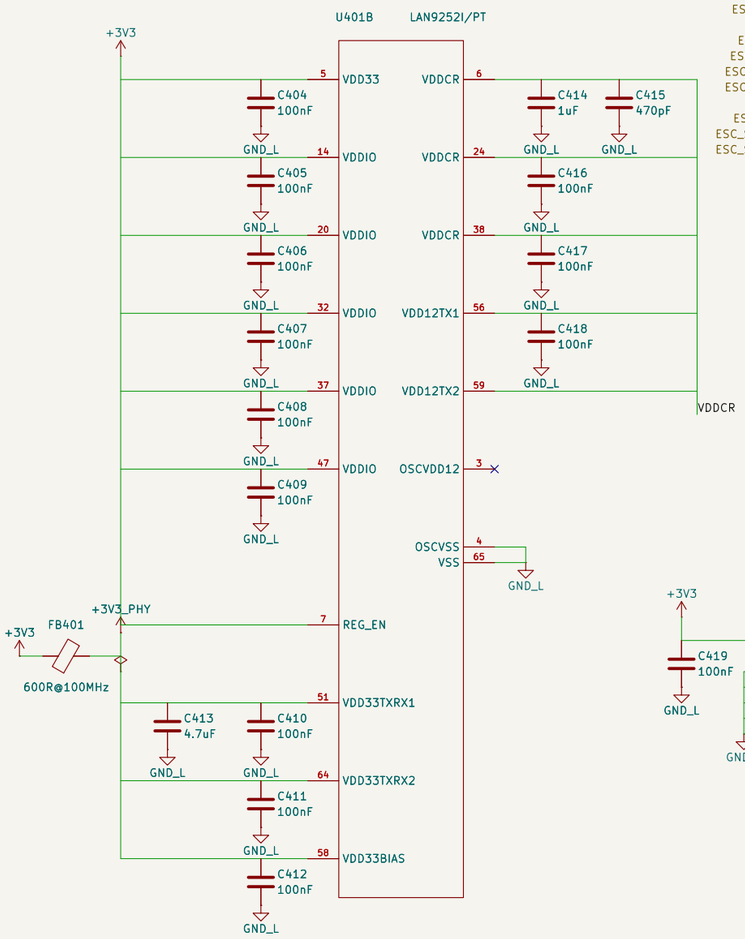
Title: LCEC-IO — 3 MCU

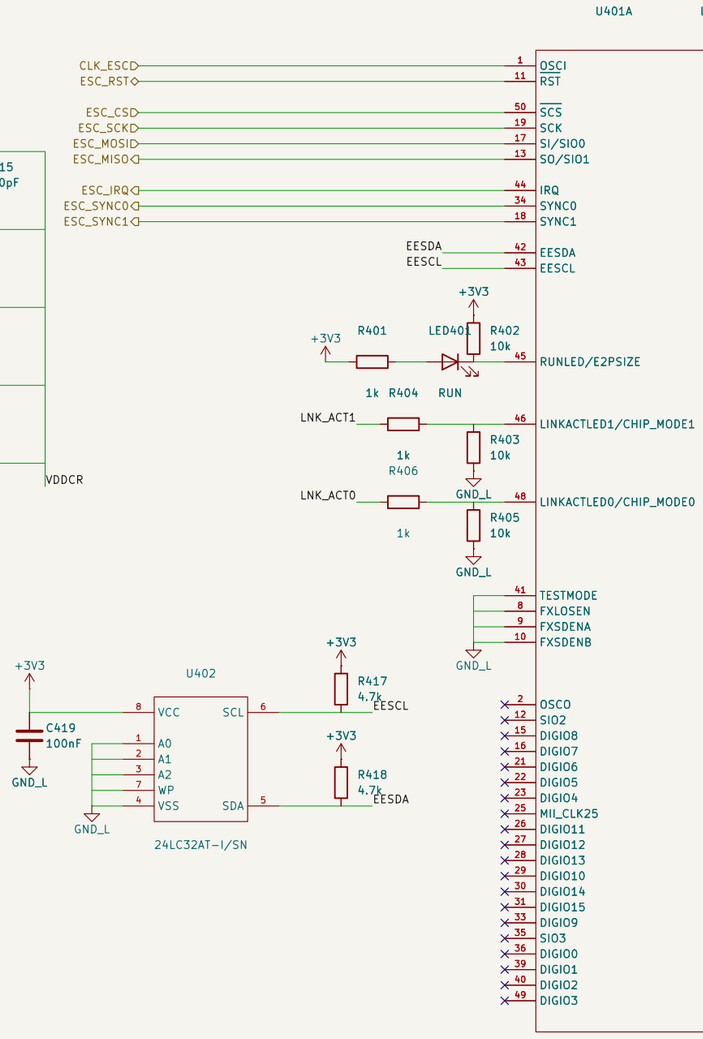
Size: A3	Date: 2026-11-06	**Rev: A**
KiCad E.D.A. 10.0.8		Id: 3/4

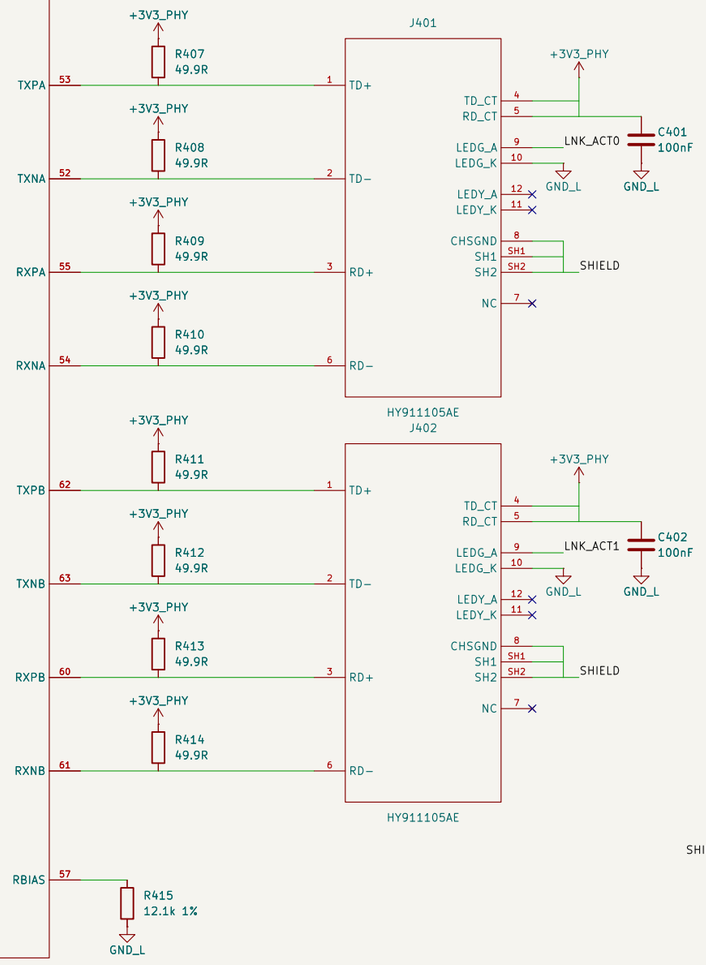
4 ETHERCAT – LAN9252 (SPI), ESI EEPROM, two RJ45 jacks with magnetics, RUN and ERR LEDs

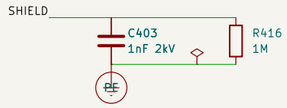
Chip mode 00 = 2 ports (PHY A = port 0 = J401, PHY B = port 1 = J402). The PDI is chosen by the ESI EEPROM content (SPI slave). Rows and values follow the LAN9252 datasheet DS00001909A (sections 3, 4, 7, 12.10 and 14).











Design notes

RUNLED / E2PSIZE: 10 k pull-up = EEPROM size strap high (32 Kbit and up, 2-byte addressing). The pin is open drain, so the RUN LED is +3V3 > 1 k > LED > pin.

LINKACTLED0 / CHIP_MODE0 and LINKACTLED1 / CHIP_MODE1: 10 k pull-downs = chip mode 00 (2 ports) and active-high LED outputs; 1 k to the green LED in each jack.

TESTMODE to ground (required). FXLOSEN, FXSDENA and FXSDENB to ground = copper twisted pair. REG_EN to 3.3 V = internal 1.2 V regulators. OSCO open (single-ended oscillator on OSCI).

Pins 12, 15 to 16, 21 to 23, 25 to 31, 33, 35, 36, 39, 40 and 49 (host bus, MII, digital I/O) are unused in SPI mode and left open.

Each twisted-pair line has 49.9 R to +3V3_PHY (ferrite-filtered 3.3 V); the magnetics centre taps are on the same net. RBIAS 12.1 k 1 % to ground.

VDDCR is the regulator output: 1 uF + 470 pF at pin 6. VDD12TX1 and VDD12TX2 are tied to it. The ESI EEPROM must hold PDI = SPI before the first run.

LAN9252 in SPI mode, 24LC32A ESI EEPROM, two RJ45 jacks, RUN and ERR LEDs

Yantra Innovation – LCEC-IO

Sheet: /EtherCAT/

File: 04_ethercat.kicad_sch

Title: LCEC-IO – 4 EtherCAT

Size: A3	Date: 2026-11-06	**Rev: A**
KiCad E.D.A. 10.0.8		Id: 4/4